II. Dedekind Domains factorization 1. Some CA review

Def II.1.1. B be the integral domain, $K = \mathrm{Frac}(B)$. B is a valuation ring of K iff $\forall x \neq 0$ $x \in B$ or $x^{-1} \in B$

Def II 1.2 Let K be the field. A discrete valuation on K is a mapping $v : K^* \to \mathbb{Z}$ such that (1) $v(xy) = v(x) + v(y)$

(2) $v(x+y) \geq \min\{v(x), v(y)\}$.

Def II.1.3. The valuation ring of v is the set consisting 0 and all $x \in K^*$ s.t. $v(x) \geq 0$

An integral domain A is a discrete valuation ring iff A is the valuation ring of v in $\mathrm{Frac}(A)$.

Def II.1.4. Let A be the Noetherian domain of dimension one (Sup{length of the prime ideal chain}), if A is integrally closed, then A is called the Dedekind domain

Examples II.1.5. (1). $\mathbb{Z}_{(p)} = \left\{ \frac{m}{n} \middle| (n,p) = 1 \right\}$. is the discrete valuation ring, with valuation

$v : \mathbb{Q}^\times \to \mathbb{Z}$. $\frac{m}{n}$ $(\gcd(m,n)=1) \mapsto v_p(\frac{m}{n})$.

where we can find that: ① $\mathbb{Z}_{(p)}$ has only one exactly prime ideal (p).

② one prime element (up to associats) p.

(2) Let A be PID, $K = \mathrm{Frac}(A)$, π be the prime element of A.

$\forall c \in K^\times$, c can be expressed uniquely in the form $c = \pi^m \frac{a}{b}$

$(a, \pi) = (b, \pi) = 1$. $v(c) = m$.

Let check (2). ① $\forall c \in K^\times$, say $c = \frac{x}{y}$. $x \in A$. $y \in A \setminus \{0\}$. $x = \pi^m a$, $y = \pi^n b$

$\Rightarrow c = \frac{x}{y} = \pi^{m-n} \frac{a}{b}$

② $v(x+y) := v(\pi^m \frac{a}{b} + \pi^n \cdot \frac{c}{d})$ $(m > n)$, a, b, c, d are coprime to π

$= n$ $(v(\pi a + b) = 0)$ □.

Thm II.1.6. (Unique factorization theorems) Let A be a Noetherian domain of dimension 1. Then every non-zero ideal $\mathfrak{a}$ in A can be uniquely expressed as a product of ~~prime~~ primary ideals whose radicals are all distinct.

proof: Since A is Noetherian, $\mathfrak{a}$ has a minimal primary decomposition $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$.

$\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary.

By $\dim A = 1$, we have: $\mathfrak{p}_i$ maximal. $\Rightarrow$ $\mathfrak{p}_i$ are coprime

Thus by CRT, $\bigcap_{i=1}^n \mathfrak{q}_i = \prod_{i=1}^n \mathfrak{q}_i$; Uniqueness: $\mathfrak{a} = \prod \mathfrak{q}_i = \bigcap_{i=1}^n \mathfrak{q}_i$ By Thm VI.1.11. □

It's unique

Lemma II.1.7. (Basis properties of the valuation ring) A be the valuation ring of $\mathrm{Frac}(A)=K$, then

(1) A is a local ring

(2) A is integrally closed in K

proof: (1) Use the fact that: $\mathfrak{m}$ is the ideal of A s.t. $\forall x\in A\setminus\mathfrak{m}$, x is a unit

then A is the local ring

(2) Obvious $\square$.

(3) A has exactly one prime ideal

Prop II.1.7$\frac{1}{2}$. (Basis property of DVR) An ID A is a DVR iff (1) A is Noetherian; (2) A is integrally closed,

Prop II.1.8 Let A be a Noetherian local domain of dimension one, $\mathfrak{m}$ its maximal ideal, k be the residue field. Then TFAE:

i) A is a DVR

ii) A is integrally closed.

iii) $\mathfrak{m}$ is a principal ideal.

iv) $\dim_k(\mathfrak{m}/\mathfrak{m}^2)=1$.

v) Every non-zero ideal is a power of $\mathfrak{m}$

vi) $\exists x\in A$ s.t. every non-zero ideal is of the form (x^k), $k\geq 0$.

Thm II.1.9. Let A be the Noetherian domain with $\dim A=1$. Then TFAE:

(1) A is integrally closed;

(2) Every primary ideal in A is a prime power

(3) Every local ring $A_{\mathfrak{p}}$ $(\mathfrak{p}\neq 0)$ is a DVR.

proof: i) $\Rightarrow$ ii)

From Lemma II.1.7. (2)

ii) $\Rightarrow$ iii) A is Noetherian local $\Rightarrow \forall a\in\mathfrak{m}$ (a) is $\mathfrak{m}$-primary

$\sqrt{(a)}=\mathfrak{m} \Rightarrow \exists n$. s.t. $\mathfrak{m}^n\subseteq(a)$ $\mathfrak{m}^{n-1}\not\subseteq(a)$.

$\Rightarrow \exists b\in\mathfrak{m}^{n-1}$. $b\notin(a)$ Let $x=\frac{a}{b}$

Then $x^{-1}\in A$. Thus x^{-1} not integral over A

By previous thm, $x^{-1}\mathfrak{m}\not\subseteq\mathfrak{m}$. But $x^{-1}\mathfrak{m}=\frac{b}{a}\mathfrak{m}\subseteq A$

By $\mathfrak{m}$ maximal, $x^{-1}\mathfrak{m}=A \Rightarrow \mathfrak{m}=(x)$

iii) $\Rightarrow$ (iv) $\mathfrak{m}/\mathfrak{m}^2\neq 0$ (Since A is Noetherian, local).

$\Rightarrow \dim_k(\mathfrak{m}/\mathfrak{m}^2)=1$ (Not so clear).

iv) $\Rightarrow$ (v) $a\neq\mathfrak{m}^n$. Take $A/\mathfrak{m}^n$ be Artinian thus a is a power of $\mathfrak{m}$

v) $\Rightarrow$ (vi) $\mathfrak{m}\neq\mathfrak{m}^2 \Rightarrow \exists x\in\mathfrak{m}$. $x\notin\mathfrak{m}^2$. by (v). $\mathfrak{m}^r=(x)$.

Thus $r=1 \Rightarrow \mathfrak{m}=(x)$. $(x^k)=\mathfrak{m}^k$

vi) $\Rightarrow$ i) Let $v: K^*\to\mathbb{Z}$ $a\mapsto v(a)$ where $(a)=(x^k)$.

$v(ab^{-1})=v(a)-v(b)$

II.1.9. proof: (1)⟺(2) $\forall \mathfrak{p} \in A$ $\sqrt{\mathfrak{q}}=\mathfrak{p}, \mathfrak{q} A_{\mathfrak{p}}$ is maximal where $A_{\mathfrak{p}}$ satisfies II 1.8 conditions
Thus $\mathfrak{p}A_{\mathfrak{p}}$ is a prime power. then $\mathfrak{q} = (\mathfrak{p}A_{\mathfrak{p}})^c$ is a prime power.

(2)⟺(3). Obvious. Since localization preserves finite products, prime ideals □

Rmk. In this prop, we can conclude that ~~every~~ the localization of the Dedekind domain is the DVR. (where we can define local dicrete value)

Thm II.1.10. (Unique factorization of ideals) In a Dedekind domain A every non-zero ideal has a unique factorization as a product of prime ideals

proof: A is Noetherian domain where $\dim A = 1$. Thus $\forall \mathfrak{a} \subseteq A$. $\mathfrak{a} = \prod_{i=1}^{n} \mathfrak{q}_i$ $\sqrt{\mathfrak{q}_i} = \mathfrak{p}_i$ (prim
$\mathfrak{q}_i = \mathfrak{p}_i^{r_i}$. Thus $\mathfrak{a} = \prod_{i=1}^{n} \mathfrak{p}_i^{r_i}$. where $\mathfrak{p}_i, r_i$ are unique ~~up to order~~ up to the factorization □

Thm II.1.11. L be the algebraic number field, then $\mathcal{O}_L$ is a Dedekind domain.

pf: Lemma: Let A be an integrally closed domain, $K = \mathrm{Frac}(A)$. L be a finite seperable algebraic extension of K, $B = \mathrm{Cl}_{A_L}(A)$. Then $\exists$ basis $v_1, \dots, v_n$ of L over K s.t. $B \subseteq \sum_{j=1}^{n} A v_j$

proof: (skipped) □

Back to the proof: L is the separable extension of $\mathbb{Q}$. ($\mathrm{char}\,\mathbb{Q} = 0$).
$\exists$ basis $v_1, \dots, v_n$ of L over $\mathbb{Q}$ s.t. $\mathcal{O}_L \subseteq \sum_{j=1}^{n} \mathbb{Z} v_j$.

$\Rightarrow$ ① $\mathcal{O}_L$ is finitely generated as a $\mathbb{Z}$-module, hence Noetherian

② $\mathcal{O}_L$ is integrally-closed.

③. $\dim \mathcal{O}_L = 1$: $\forall \mathfrak{p} \subseteq \mathcal{O}_L$ prime we know that $\mathfrak{p}^c = \mathfrak{p} \cap \mathbb{Z} \neq 0$. ($\mathbb{Z}$ is PID) is maximal
$\Rightarrow$ $\mathfrak{p}$ maximal

Therefore, $\mathcal{O}_L$ is ~~a~~ Dedekind domain □.

The ideal class group

Def II.2.1. A be the ~~ring~~ Dedekind domain, a fractional ideal of A is a non-zero A-submodule $\mathfrak{a}$ of K, s.t. $d\mathfrak{a} = \{da \mid a \in \mathfrak{a}\}$ for some $d \in A \setminus \{0\}$ (or K). ($K = \mathrm{Frac}\, A$)

Def II.2.2. (1) An A-submodule M of K is an invertible ideal if $\exists$ submodule N of K s.t. $MN = K$

Every nonzero element b of K defines a fractional ideal

$(b) := \{ba \mid a \in A\}$. , called the principal fractional ideal.

(2)

Example II.2.3. Let A be the DVR. with maximal ideal $\mathfrak{p}$ and $K = \mathrm{Frac}(A)$. π be the generator of $\mathfrak{p}$. Then every nonzero element of K can be written uniquely $a = u\pi^n$. u is a unit in A, $n \in \mathbb{Z}$

Let $\mathfrak{a}$ be a fractional ideal of A. Then $d\mathfrak{a} \subseteq A$ for some $d \in A$.

W.L.O.G Suppose that $d = \pi^n \Rightarrow \mathfrak{a} = (\pi^{m-n})$.

Hence the fractional ideals of A are of the form (π^m). $\quad m \mapsto (\pi^m), \mathbb{Z} \to \mathrm{Id}(A)$

Thm II.2.4 Let A be a Dedekind domain. The set $\mathrm{Id}(A)$ of fractional ideals is the free Abelian group on the set of nonzero prime ideals.

Lemma: (1) Let A be a local domain, then A is a DVR. $\Leftrightarrow$ Every non-zero fractional ideal is invertibal

(2) Let A be an integral domain, then A is a Dedekind domain $\Leftrightarrow$ Every non-zero fractional ideal is invertible

pf: (skipped)

Back to the proof: It's trivial since $\mathrm{Id}(A)$ is the set of invertible ideals, thus free Abelian group

Let K be a Dedekind domain

Rmk: $h_K = 1 \Leftrightarrow$ (~~Reg~~ PID $\Leftrightarrow$ UFD).

Def II.2.5. The ideal class group $\mathrm{Cl}(A)$ of A is $:= \mathrm{Id}(A)/P(A)$ where $P(A)$ is the subgroup of principal ideals. The class number of A is the order of $\mathrm{Cl}(A)$

In particular, K be the algebraic number field. $\mathcal{O}_K = A$. Then h_K is the class number of $\mathcal{O}_K$